

Project: Navier-Stokes POD vs PODNN

Let us consider the two-dimensional spatial domain $\Omega = (0, 1)^2$. We want to solve the following parametrized problem: given $\boldsymbol{\mu} = (\mu_0, \mu_1) \in \mathcal{P} = [0.1, 10] \times [1.0, 3.0]$, find $u(\boldsymbol{\mu})$ such that

$$\begin{cases} -\mu_0 \nabla \cdot (\nabla \mathbf{u}) + (\nabla \mathbf{u}) \mathbf{u} + \nabla p = \mathbf{f}(\mathbf{x}; \mu_1) & \text{in } \Omega \\ (\nabla \cdot \mathbf{u}) = 0 & \text{in } \Omega \\ u = 0 & \text{in } \partial\Omega \end{cases}$$

and the pressure constant is fixed imposing $p = 0$ in the vertex $(0.0, 0.0)$.

The source term defined as

$$\mathbf{f}(\mathbf{x}; \mu_1) = -(\mu_1^3 \pi^2 \cos(\mu_1^2 \pi x) - \mu_1^2 \pi^2) \sin(\mu_1 \pi y) \cos(\mu_1 \pi y) + (\mu_1 \pi \cos(\mu_1 \pi x) \cos(\mu_1 \pi y))$$

Tasks:

1. solve the problem by means of POD-Galerkin method over a Finite Element full order model
2. solve the problem with a parametric PODNN
3. compare the two approaches in terms of computational costs and accuracy with respect to the full order model
4. *Optional:* solve the problem with the PINN approach and compare it to the other two strategies

Remember that in PINNs...

Given a neural network $\tilde{w}(\mathbf{x}, \boldsymbol{\mu})$, the loss will be of the form

$$MSE \doteq MSE_b^\mu + \lambda MSE_p^\mu.$$

where the *boundary* MSE is:

$$MSE_b^\mu \doteq \frac{1}{N_b} \sum_{k=1}^{N_b} |\tilde{w}(\mathbf{x}_k^b, \boldsymbol{\mu}_k^b) - u(\mathbf{x}_k^b, \boldsymbol{\mu}_k^b)|^2,$$

for $(\mathbf{x}_k^b, \boldsymbol{\mu}_k^b) \in \partial\Omega \times \mathcal{P}$. While the *physical* MSE is

$$MSE_p^\mu \doteq \frac{1}{N_p} \sum_{k=1}^{N_p} |\mathcal{R}(\tilde{w}(\mathbf{x}_k^p, \boldsymbol{\mu}_k^p))|^2,$$

where $(\mathbf{x}_k^p, \boldsymbol{\mu}_k^p) \in \Omega \times \mathcal{P}$. Finally, λ is the hyperparameter that can be used to balance the two different contributions.